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Restaurant Customer Satisfaction

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
```

```
df = pd.read_csv("4.tips.csv")
df
```

	total_bill	tip	gender	smoker	day	time	size	
0	16.99	1.01	Female	No	Sun	Dinner	2	
1	10.34	1.66	Male	No	Sun	Dinner	3	
2	21.01	3.50	Male	No	Sun	Dinner	3	
3	23.68	3.31	Male	No	Sun	Dinner	2	
4	24.59	3.61	Female	No	Sun	Dinner	4	
...	...	...	...	...	...	...	...	
239	29.03	5.92	Male	No	Sat	Dinner	3	
240	27.18	2.00	Female	Yes	Sat	Dinner	2	
241	22.67	2.00	Male	Yes	Sat	Dinner	2	
242	17.82	1.75	Male	No	Sat	Dinner	2	
243	18.78	3.00	Female	No	Thur	Dinner	2	

244 rows × 7 columns

Next steps:

New interactive sheet

```
df.head()
```

	total_bill	tip	gender	smoker	day	time	size	
0	16.99	1.01	Female	No	Sun	Dinner	2	
1	10.34	1.66	Male	No	Sun	Dinner	3	
2	21.01	3.50	Male	No	Sun	Dinner	3	
3	23.68	3.31	Male	No	Sun	Dinner	2	
4	24.59	3.61	Female	No	Sun	Dinner	4	

Next steps:

New interactive sheet

```
df.columns
```

```
Index(['total_bill', 'tip', 'gender', 'smoker', 'day', 'time', 'size'], dtype='object')
```

1.NUMPY ARRAYS:

Convert the 'Tip_Amount' column into a NumPy array. Calculate the total tip revenue.

```
tips_array = df['tip'].to_numpy()
print(tips_array)
```

```
[ 1.01  1.66  3.5   3.31  3.61  4.71  2.    3.12  1.96  3.23  1.71  5.
  1.57  3.    3.02  3.92  1.67  3.71  3.5   3.35  4.08  2.75  2.23  7.58
  3.18  2.34  2.    2.    4.3   3.    1.45  2.5   3.    2.45  3.27  3.6
  2.    3.07  2.31  5.    2.24  2.54  3.06  1.32  5.6   3.    5.    6.
  2.05  3.    2.5   2.6   5.2   1.56  4.34  3.51  3.    1.5   1.76  6.73
  3.21  2.    1.98  3.76  2.64  3.15  2.47  1.    2.01  2.09  1.97  3.
  3.14  5.    2.2   1.25  3.08  4.    3.    2.71  3.    3.4   1.83  5.
  2.03  5.17  2.    4.    5.85  3.    3.    3.5   1.    4.3   3.25  4.73
  4.    1.5   3.    1.5   2.5   3.    2.5   3.48  4.08  1.64  4.06  4.29
  3.76  4.    3.    1.    4.    2.55  4.    3.5   5.07  1.5   1.8   2.92
  2.31  1.68  2.5   2.    2.52  4.2   1.48  2.    2.    2.18  1.5   2.83
  1.5   2.    3.25  1.25  2.    2.    2.    2.75  3.5   6.7   5.    5.
  2.3   1.5   1.36  1.63  1.73  2.    2.5   2.    2.74  2.    2.    5.14
  5.    3.75  2.61  2.    3.5   2.5   2.    2.    3.    3.48  2.24  4.5
  1.61  2.    10.   3.16  5.15  3.18  4.    3.11  2.    2.    4.    3.55
  3.68  5.65  3.5   6.5   3.    5.    3.5   2.    3.5   4.    1.5   4.19
  2.56  2.02  4.    1.44  2.    5.    2.    2.    4.    2.01  2.    2.5
  4.    3.23  3.41  3.    2.03  2.23  2.    5.16  9.    2.5   6.5   1.1
  3.    1.5   1.44  3.09  2.2   3.48  1.92  3.    1.58  2.5   2.    3.
  2.72  2.88  2.    3.    3.39  1.47  3.    1.25  1.    1.17  4.67  5.92
  2.    2.    1.75  3.    ]
```

```
total_tips=tips_array.sum()
print("Total Tips:",total_tips)
```

```
Total Tips: 731.5799999999999
```

2.PANDAS DATAFRAMES

Load the dataset. Filter for transactions that occurred on a weekend and for a 'Party_Size' greater than 4.

```
filtered_transaction=df[(df['day'].isin(['Sat','Sun'])) & (df['size']>4)]
filtered_transaction
```

	total_bill	tip	gender	smoker	day	time	size	
155	29.85	5.14	Female	No	Sun	Dinner	5	
156	48.17	5.00	Male	No	Sun	Dinner	6	
185	20.69	5.00	Male	No	Sun	Dinner	5	
187	30.46	2.00	Male	Yes	Sun	Dinner	5	
216	28.15	3.00	Male	Yes	Sat	Dinner	5	

Next steps: [New interactive sheet](#)

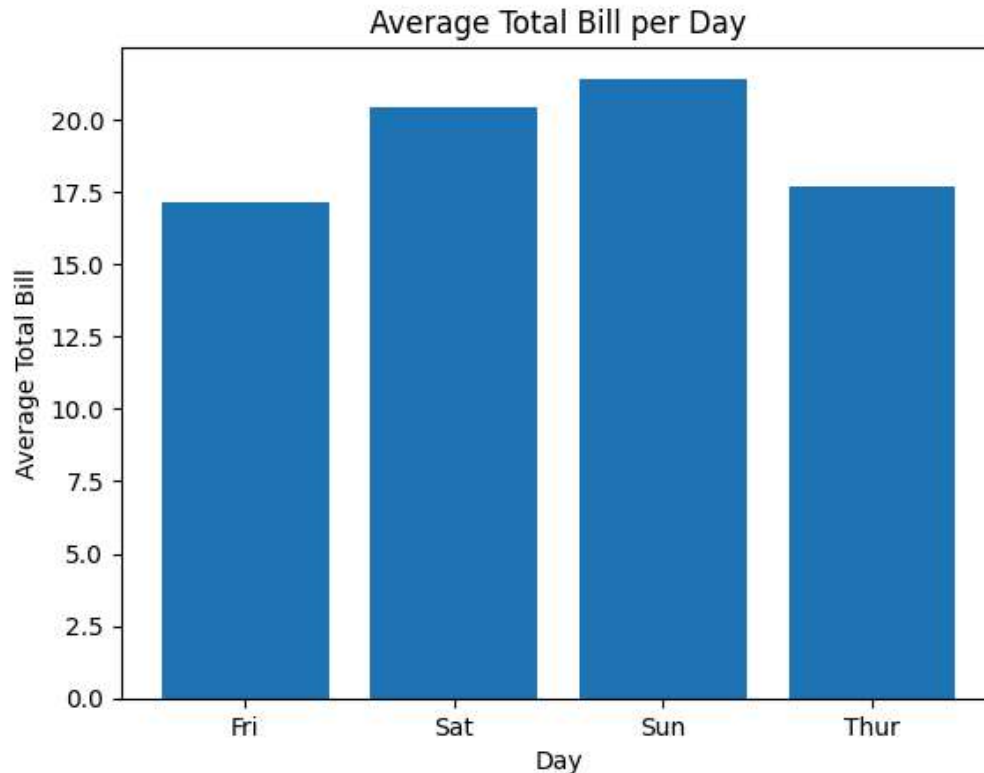
3.MATPLOTLIB PLOTS:

Create a bar chart showing the average 'Total_Bill' for each 'Day_of_Week'.

```
avg = df.groupby('day')['total_bill'].mean()
plt.bar(avg.index, avg.values)

plt.title("Average Total Bill per Day")
plt.xlabel("Day")
plt.ylabel("Average Total Bill")

plt.show()
```



4.FREQUENCY DISTRIBUTION:

What is the frequency distribution of the 'Party_Size' at each table.

```
df['size'].value_counts()
```

	count
size	
2	156
3	38
4	37
5	5
1	4
6	4

dtype: int64

5.AVERAGES:

Calculate the mean and median of the 'Total_Bill' amounts.

```
mean = df['total_bill'].mean()
median = df['total_bill'].median()

print("Mean of the Total Bill:",mean)
print("Median of the Total Bill:",median)
```

```
Mean of the Total Bill: 19.78594262295082
Median of the Total Bill: 17.795
```

6.VARIABILITY:

Find the standard deviation of the 'Tip_Amount' to measure how consistent tips are.

```
std_tip = df['tip'].std()
print("Standard Deviation of Tip:", std_tip)
```

```
Standard Deviation of Tip: 1.3836381890011826
```

7.NORMAL CURVES:

Plot a histogram of the 'Total_Bill' and a normal curve with the same mean and standard deviation.

```
from scipy.stats import norm

plt.hist(df['total_bill'], density=True)

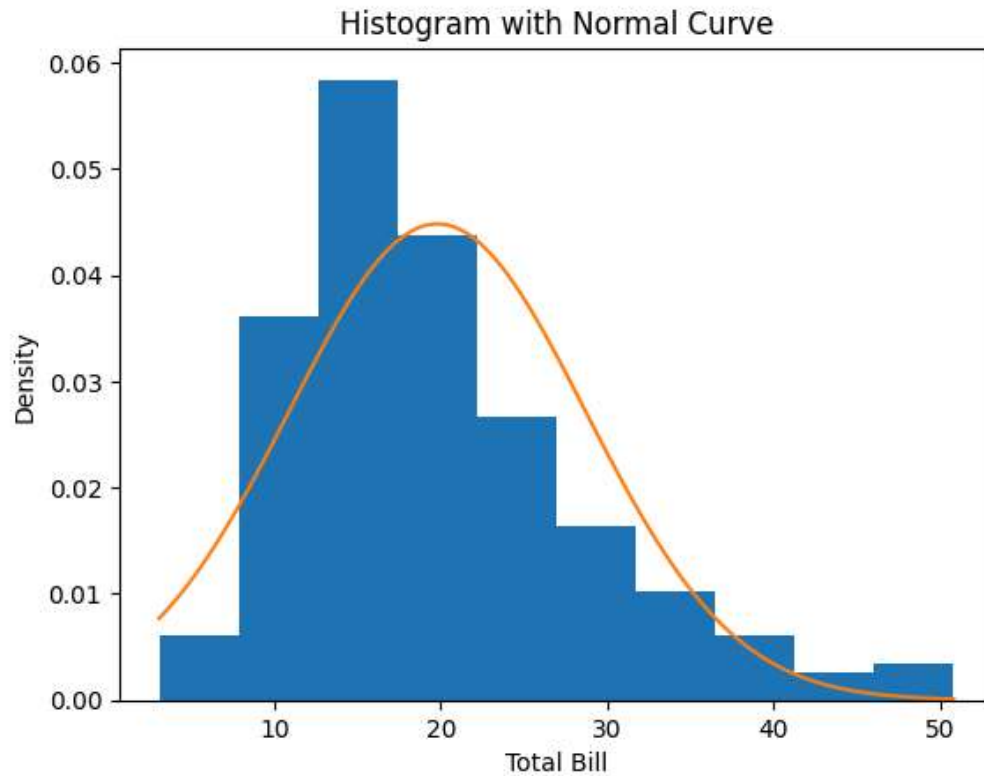
m = df['total_bill'].mean()
```

```
s = df['total_bill'].std()

x = np.linspace(df['total_bill'].min(), df['total_bill'].max(), 100)
plt.plot(x, norm.pdf(x, m, s))

plt.title("Histogram with Normal Curve")
plt.xlabel("Total Bill")
plt.ylabel("Density")

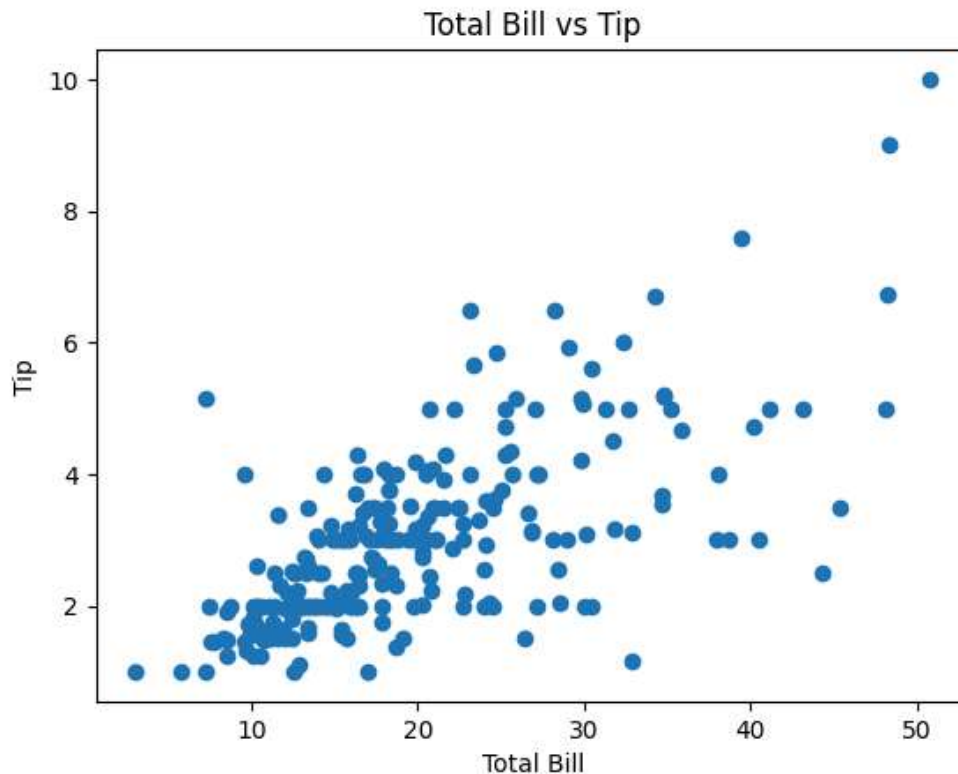
plt.show()
```



8.CORRELATION & SCATTER PLOTS:

Create a scatter plot to visualize the relationship between 'Total_Bill' and 'Tip_Amount'.

```
plt.scatter(df['total_bill'],df['tip'])
plt.xlabel('Total Bill')
plt.ylabel('Tip')
plt.title('Total Bill vs Tip')
plt.show()
```



9.CORRELATION COEFFICIENT:

Calculate the correlation coefficient between 'Total_Bill' and 'Tip_Amount'.

```
corr=(df['total_bill'].corr(df['tip']))  
print("Correlation Coefficient:", corr)
```

Correlation Coefficient: 0.6757341092113641

10.REGRESSION

Perform a linear regression to predict 'Tip_Amount' from the 'Total_Bill'. Plot the regression line and use the model to predict the tip for a bill of \$50.

```
X = df[['total_bill']]  
y = df['tip']  
  
model = LinearRegression()  
model.fit(X, y)  
  
pred = model.predict([[50]])  
print("Predicted Tip for $50 bill:", pred)  
  
plt.scatter(X, y)  
plt.plot(X, model.predict(X))  
  
plt.title("Total Bill vs Tip")  
plt.xlabel("Total Bill")
```

```
Predicted Tip for $50 bill: [6.17149548]
/usr/local/lib/python3.12/dist-packages/sklearn/utils/validation.py:2739: UserWarning: X does
  warnings.warn(
```

